

Model Equivalence for Linear Non-Gaussian Causal Models Under General Confounding

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Abstract

We study the structure identification problem for linear non-Gaussian causal models that allow for general (also non-linear) latent confounding. We focus on models without feedback loops, which are naturally represented by acyclic directed mixed graphs (ADMGs). In this setting, two ADMGs are said to be model equivalent if they induce the same set of probability distributions. Our main result is an algebraic condition that is necessary and sufficient for model equivalence of two ADMGs. The condition amounts to solving systems of quadratic equations and is computationally tractable for challenging cases. In addition, we give a graphical characterization of model equivalence in the case of removal of a directed edge. Finally, we derive a condition on the bidirected components of two ADMGs that is necessary for model equivalence.

1 INTRODUCTION

Owing to their computational and theoretical tractability, *linear structural causal models* (SCMs) play an important role in causal inference [Pearl, 2013]. Indeed, a vast literature on linear SCMs provides refined results on identification and estimation of causal effects and causal structure; see the review of Drton [2018] that focuses on covariance structure and Gaussian distributions, or the monograph of Shimizu [2022] that treats linear non-Gaussian models. A central challenge is to gain an understanding of how latent confounding manifests itself in constraints on observable distributions [e.g., Shpitser et al., 2018, Drton et al., 2020] and how it impacts effect and structure identifiability [e.g., Barber et al., 2022, van Ommen, 2024, Dong et al., 2024].

In this paper, we consider linear non-Gaussian SCMs with latent confounding, under the common assumption that the modeled system is acyclic (i.e., without feedback loops).

When latent effects are linear, a connection to overcomplete independent component analysis (ICA) has facilitated numerous advances; recent examples include Salehkaleybar et al. [2020], Tramontano et al. [2024b]. Our interest, however, is in the less explored setting of models that permit general (non-linear) confounding [Wang and Drton, 2023, Tramontano et al., 2024a]. Such models may be compactly represented using acyclic directed mixed graphs (ADMGs) [Richardson et al., 2023, Zhao, 2025], where directed edges encode direct causal effects among observed variables and bidirected edges indicate possible latent confounding. The mixed graph perspective is rooted in the notion of latent projection [Verma and Pearl, 1990], in which consideration of explicit latent variables is relaxed to mere dependencies among noise terms in structural equations.

Let $X = (X_v)_{v \in V}$ be a (finite) collection of observed random variables, and let $G = (V, E_{\rightarrow}, E_{\leftrightarrow})$ be an ADMG with vertex set V , set of directed edges $E_{\rightarrow}$, and set of bidirected edges $E_{\leftrightarrow}$. A linear SCM given by G postulates that each variable is a linear function of parent variables and noise:

$$X_v = \sum_{w: w \rightarrow v \in E_{\rightarrow}} \lambda_{wv} X_w + \varepsilon_v, \quad v \in V.$$

The $\lambda_{wv} \in \mathbb{R}$ are the direct causal effects, and the ε_v are error terms, which we will assume to be non-Gaussian. As we will make more precise later, whenever there is a bidirected edge $v \leftrightarrow w \in E_{\leftrightarrow}$, then ε_v and ε_w are allowed to be dependent; the dependence stems from a latent common cause affecting both X_v and X_w (in possibly nonlinear ways). In applications, the effects (λ_{wv}) and the joint distribution of $\varepsilon = (\varepsilon_v)_{v \in V}$ are unknown and have to be estimated.

Example 1.1. Under the model given by the ADMG in Figure 1, we have

$$\begin{aligned} X_1 &= \varepsilon_1, & X_2 &= \lambda_{12}X_1 + \lambda_{32}X_3 + \varepsilon_2, \\ X_3 &= \varepsilon_3, & X_4 &= \lambda_{34}X_3 + \lambda_{24}X_2 + \varepsilon_4. \end{aligned}$$

The graph allows for the dependences $\varepsilon_1 \not\perp\!\!\!\perp \varepsilon_2$ and $\varepsilon_2 \not\perp\!\!\!\perp \varepsilon_4$.

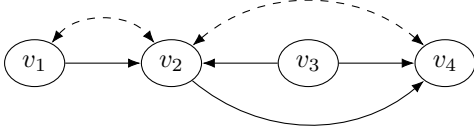


Figure 1: An example of an ADMG with four nodes.

In practice, one often has to learn the underlying graph from data. This model selection problem is commonly referred to as *causal discovery* [Spirtes and Zhang, 2019]. For linear non-Gaussian models that are acyclic and without latent confounding (i.e., $E_{\leftrightarrow} = \emptyset$), ICA results imply that the graph, now a directed acyclic graph (DAG), can be uniquely identified from the observational distribution of X [Shimizu, 2022]. Wang and Drton [2023] show that the ADMG of a linear non-Gaussian model is also uniquely identified under the restriction that it be *bow-free*. This means that between any two vertices there may be at most one edge, i.e., a *bow* with both $v \rightarrow w$ and $v \leftrightarrow w$ is not allowed. Such graphs are sometimes termed *simple* [Améndola et al., 2020].

As we show in this paper, general ADMGs with bows may induce identical or statistically indistinguishable linear non-Gaussian SCMs. This model equivalence of ADMGs is the core problem we study, and we obtain the following main contributions:

- We give an algebraic condition that is necessary and sufficient for model equivalence (Theorem 3.1). This characterization reduces the problem to solving systems of quadratic equations and admits a practically feasible implementation in computer algebra systems.
- We provide a graphical characterization for determining whether the removal of a directed edge from an ADMG changes the associated set of observational distributions (Theorem 3.2).
- We prove that if two ADMGs are model equivalent, then their bidirected parts must satisfy certain connectedness constraints (Theorem 3.4). In particular, they must have the same number of bidirected edges.
- Building on these results, we develop an algorithm for checking model equivalence (Algorithm 2), which we implement and evaluate experimentally in Section 4.

2 MODEL DEFINITION

2.1 GRAPHICAL CONCEPTS

Formally, a mixed graph is a triple $G = (V, E_{\rightarrow}, E_{\leftrightarrow})$, where $E_{\rightarrow}, E_{\leftrightarrow} \subseteq V \times V \setminus \{(v, v) : v \in V\}$ denote the set of directed and bidirected edges, respectively. The directed part of G is the directed graph $G_D = (V, E_{\rightarrow})$, and the bidirected part is the graph $G_B = (V, E_{\leftrightarrow})$. We write $p := |V|$ for the number of nodes of G .

Let $u, v \in V$ be two vertices in G . A directed (or bidirected) path from u to v is a sequence of nodes $\pi = (v_0, \dots, v_k)$ with $v_0 = u$ and $v_k = v$ such that $v_i \rightarrow v_{i+1} \in E_{\rightarrow}$ (or $v_i \leftrightarrow v_{i+1} \in E_{\leftrightarrow}$) for all $i = 0, \dots, k-1$. We define $\mathcal{P}(u, v)$ to be the set of all directed paths from u to v . We say that a path is trivial if $\pi = u$ contains only a single node and no edge. A directed cycle is a non-trivial directed path from a node u to itself. If a graph contains no directed cycle, then it is acyclic. We refer to this class of mixed graphs as *acyclic directed mixed graphs (ADMGs)*. Figure 1 provides an example.

We say that a subset $C \subseteq V$ is *connected* in G_B if for every pair $u, v \in C$, there is a bidirected path π in G_B that consists only of vertices in C . For example, if G is the graph from Figure 1, then the set $C = \{1, 2, 4\}$ is connected in G_B while $C = \{1, 4\}$ is not.

The following genealogical relations of vertices are commonly used:

- (parents) $\text{pa}(v) := \{u \in V : u \rightarrow v \in G\}$,
- (children) $\text{ch}(v) := \{u \in V : v \rightarrow u \in G\}$,
- (ancestors) $\text{an}(v) := \{u \in V : \mathcal{P}(u, v) \neq \emptyset \in G\}$,
- (descendants) $\text{de}(v) := \{u \in V : \mathcal{P}(v, u) \neq \emptyset \in G\}$,
- (siblings) $\text{sib}(v) := \{u \in V : v \leftrightarrow u \in G\}$,
- $\text{Sib}(v) := \text{sib}(v) \cup \{v\}$.

Note that in our setting we have $v \in \text{an}(v)$ and $v \in \text{de}(v)$, via trivial paths. For a subset $U \subseteq V$, we define $\text{pa}(U) = \bigcup_{u \in U} \text{pa}(u)$ and make the analogous convention for the other relations.

Let $U = \{u_1, \dots, u_n\}$ and $W = \{w_1, \dots, w_n\}$ be two subsets of V with the same cardinality n . We say that $\Pi = (\pi_1, \dots, \pi_n)$ is a system of paths between U and W , if there exists a permutation $\sigma_\Pi \in S_n$ such that $\pi_k \in \mathcal{P}(u_k, w_{\sigma_\Pi(k)})$ for all $k \in \{1, \dots, n\}$. We denote the set of all such systems by $\mathcal{P}(U, W)$. A system $\Pi \in \mathcal{P}(U, W)$ is called *non-intersecting* if $\pi_k \cap \pi_l = \emptyset$ for all $k \neq l$; here, we identified a path with the set of its nodes. The set of all non-intersecting systems in $\mathcal{P}(U, W)$ is denoted by $\hat{\mathcal{P}}(U, W)$.

2.2 LINEAR MIXED GRAPHICAL MODELS

Given an ADMG G and a fixed probability space, we define $\mathbb{R}^{G_D}$ to be the set of all matrices $\Lambda = (\lambda_{uv}) \in \mathbb{R}^{p \times p}$ such that $\lambda_{uv} = 0$ if $u \rightarrow v \notin E_{\rightarrow}$. Note that if G is acyclic, the matrix $I - \Lambda$ is invertible for all $\Lambda \in \mathbb{R}^{G_D}$, where I is the identity matrix.

A noise random vector $\varepsilon = (\varepsilon_v)_{v \in V}$ satisfies the *connected set Markov property* with respect to G_B [Drton and Richardson, 2008, Richardson, 2003] if for all non-empty subsets $C \subset V$, we have

$$\varepsilon_C \perp\!\!\!\perp \varepsilon_{V \setminus \text{Sib}(C)} \text{ whenever } C \text{ is connected in } G_B.$$

Following Tramontano et al. [2024a], we say that the noise vector ε is *generic* if for arbitrary $a, b \in \mathbb{R}^p$, it holds that

$$\begin{aligned} a^T \varepsilon \perp b^T \varepsilon &\implies \\ a_u \cdot b_v &= 0 \quad \text{for all } u = v \text{ and } u \leftrightarrow v \in E_{\leftrightarrow}. \end{aligned} \quad (2.1)$$

We write $\mathcal{M}(G_B)$ for the set of all random vectors ε that satisfy the connected set Markov property and the genericity assumption in Equation (2.1).

The genericity assumption is natural in view of the Darmois-Skitovich theorem [Darmois, 1953, Skitovich, 1954], which in our setting amounts to the fact that for $E_{\leftrightarrow} = \emptyset$, Equation (2.1) is true provided at most one of the random variables ε_v is Gaussian. The idea behind the generalization to $E_{\leftrightarrow} \neq \emptyset$ is to prohibit the possibility of linearly disentangling a general dependence between two errors ε_v and ε_w . Importantly, it can be shown that (2.1) is satisfied by general distributions. In particular, when generating the dependence in ε by linear latent factor models then the set of factor loadings for which (2.1) fails has Lebesgue measure zero. Similarly, when considering distributions with finite moments up to a suitable large order, only a null set of moments leads to failure of (2.1); we refer to Tramontano et al. [2024a, §5] for further details.

With the above preparation, we formally define the considered linear SCMs as follows:

Definition 2.1. *The linear SCM $\mathcal{M}(G)$ given by an ADMG G is the set of all p -variate real random vectors X that solve the equation system*

$$\begin{aligned} X &= \Lambda^T X + \varepsilon \\ \iff X &= (I - \Lambda)^{-T} \varepsilon = B_\Lambda \varepsilon, \end{aligned}$$

where $B_\Lambda := (I - \Lambda)^{-T}$, $\Lambda \in \mathbb{R}^{G_D}$, and $\varepsilon \in \mathcal{M}(G_B)$. The model $\mathcal{M}(G)$ is thus parametrized by the map

$$\begin{aligned} \Phi_G : \mathbb{R}^{G_D} \times \mathcal{M}(G_B) &\rightarrow \mathcal{M}(G) \\ (\Lambda, \varepsilon) &\mapsto (I - \Lambda)^{-T} \varepsilon. \end{aligned}$$

We remark that the matrix B_Λ is a *path matrix*. Its entry $(B_\Lambda)_{uv}$ is determined by the directed paths from u to v [Sullivant et al., 2010], namely,

$$(B_\Lambda)_{uv} = \sum_{\pi \in \mathcal{P}(v, u)} \prod_{w_i \rightarrow w_j \in \pi} \lambda_{ij}, \quad \forall v \in \text{an}(u) \setminus \{u\}.$$

Note that $(B_\Lambda)_{vv} = 1$ and $(B_\Lambda)_{uv} = 0$ if $v \notin \text{an}(u)$.

An important problem concerning the models we have introduced is to clarify when two different graphs yield the same model for the observed random vector X .

Definition 2.2 (Model equivalence). *Two ADMGs G and $\tilde{G}$ are model equivalent if $\mathcal{M}(G) = \mathcal{M}(\tilde{G})$.*

The results we develop below will show that, in general, only an equivalence class of graphs can be determined from the observational distribution of X .

3 MAIN RESULTS

3.1 COMPUTATIONAL ALGEBRA

Let G and $\tilde{G}$ be two arbitrary ADMGs with the same vertex set V . Then $\mathcal{M}(G) \subseteq \mathcal{M}(\tilde{G})$ if and only if for all $X = \Phi_G(\Lambda, \varepsilon) \in \mathcal{M}(G)$, there exist some $\tilde{\Lambda} \in \mathbb{R}^{\tilde{G}_D}$ and $\tilde{\varepsilon} \in \mathcal{M}(\tilde{G}_B)$ such that $X \stackrel{d}{=} (I - \tilde{\Lambda})^{-T} \tilde{\varepsilon}$, where $\stackrel{d}{=}$ denotes equal in distribution. This fact drives our first main result, an algebraic characterization of model equivalence as formalized in Definition 2.2.

Theorem 3.1. *Let G and $\tilde{G}$ be two arbitrary ADMGs with the same vertex set V . Then $\mathcal{M}(G) \subseteq \mathcal{M}(\tilde{G})$ if and only if the following system of equations has a solution in $\tilde{\Lambda} \in \mathbb{R}^{\tilde{G}_D}$ for all choices of $\Lambda \in \mathbb{R}^{G_D}$:*

$$\left\{ \begin{aligned} &((I - \tilde{\Lambda})^T B_\Lambda)_{uw_1} \cdot ((I - \tilde{\Lambda})^T B_\Lambda)_{vw_2} = 0 : \\ &(u, v) \in J^{\tilde{G}}, (w_1, w_2) \in K^G \end{aligned} \right\}.$$

Here we define

$$\begin{aligned} J^{\tilde{G}} &:= \{(u, v) : u \neq v \text{ and } u \leftrightarrow v \notin \tilde{G}\} \text{ and} \\ K^G &:= \{(w_1, w_2) : w_1 = w_2 \text{ or } w_1 \leftrightarrow w_2 \in G\}. \end{aligned}$$

According to Theorem 3.1, the model equivalence of two arbitrary ADMGs can be certified by solving systems of quadratic equations. We summarize the approach in Algorithm 1.

Algorithm 1 Check if $\mathcal{M}(G) \subseteq \mathcal{M}(\tilde{G})$.

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A ← (I − Λ̃)T(I − Λ)−T;
L ← ∅;
for each  $u \leftrightarrow v \notin \tilde{G}$ , where  $v \neq u$  do
  for each  $w_1 = w_2$  and  $w_1 \leftrightarrow w_2 \in G$  do
    L ← L ∪ {Auw1 · Avw2 = 0};
  end for
end for
if L has a solution  $\tilde{\Lambda} \in \mathbb{R}^{\tilde{G}_D}$  for all  $\Lambda \in \mathbb{R}^{G_D}$  then
  return  $\mathcal{M}(G) \subseteq \mathcal{M}(\tilde{G})$ 
else
  return  $\mathcal{M}(G) \not\subseteq \mathcal{M}(\tilde{G})$ 
end if

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Algorithm 1 requires computing the matrix $A = (I - \tilde{\Lambda})^T(I - \Lambda)^{-T}$, which takes $\mathcal{O}(p^3)$ operations. By Corollary 3.5 below, we may reject $\mathcal{M}(G) = \mathcal{M}(\tilde{G})$ if $|E_{\leftrightarrow}| \neq |\tilde{E}_{\leftrightarrow}|$. If we have $b := |E_{\leftrightarrow}| = |\tilde{E}_{\leftrightarrow}|$, then verifying model inclusion requires analyzing a system of quadratic equations L with

$$\mathcal{O}\left(\left(\frac{p^2}{2} - \frac{p}{2} - b\right) \cdot (b + p)\right)$$

many equations. Hence, the system L is small when p is small or the ADMGs have large bidirected parts.

From Theorem 3.1, one can directly show that if G and $\tilde{G}$ both have fully connected bidirected parts, then they are always model equivalent, regardless of their directed parts. Conversely, if one ADMG does not have any bidirected edges, then its equivalence class consists only of itself, i.e., it is structural identifiable.

Example 3.1. Consider the simplest possible case, where G and $\tilde{G}$ are DAGs on two nodes, $V = \{v_1, v_2\}$, with G having the single edge $v_1 \rightarrow v_2$ and $\tilde{G}$ having only the reversed edge $v_2 \rightarrow v_1$. To check whether $\mathcal{M}(G) \subseteq \mathcal{M}(\tilde{G})$, we first construct the matrix

$$A = (I - \tilde{\Lambda})^T (I - \Lambda)^{-T} = \begin{pmatrix} 1 - \tilde{\lambda}_{21}\lambda_{12} & -\tilde{\lambda}_{21} \\ \lambda_{12} & 1 \end{pmatrix}.$$

Since $v_1 \leftrightarrow v_2 \notin \tilde{G}$, we need to check if the equation system

$$\begin{aligned} L &= \{A_{11} \cdot A_{21} = 0, A_{12} \cdot A_{22} = 0\} \\ &= \{(1 - \tilde{\lambda}_{21}\lambda_{12})\lambda_{12} = 0, -\tilde{\lambda}_{21} = 0\} \end{aligned}$$

has a solution in $\tilde{\lambda}_{21} \in \mathbb{R}$ for all $\lambda_{12} \in \mathbb{R}$. Evidently, L has solution $\tilde{\lambda}_{21} = 0$ when $\lambda_{12} = 0$, but L has no solution when $\lambda_{12} \neq 0$. We may deduce that $\mathcal{M}(G) \not\subseteq \mathcal{M}(\tilde{G})$, and by symmetry, $\mathcal{M}(G) \not\supseteq \mathcal{M}(\tilde{G})$.

For a less trivial illustration of Theorem 3.1, we consider graphs related to the *instrumental variable (IV)* graph.

Example 3.2. Consider the graphs from Figure 2. Using Theorem 3.1, we may show that $\mathcal{M}(G_2) = \mathcal{M}(G_4) = \mathcal{M}(G_5)$ and that $\mathcal{M}(G_1), \mathcal{M}(G_3) \subsetneq \mathcal{M}(G_2)$. To exemplify one of the computations, consider $G := G_3$ and $\tilde{G} := G_2$. We calculate

$$A = \begin{pmatrix} 1 & 0 & 0 \\ \lambda_{13}\lambda_{32} - \tilde{\lambda}_{12} & 1 & \lambda_{32} \\ -\tilde{\lambda}_{13} + \lambda_{13} & 0 & 1 \end{pmatrix}.$$

Since $A_{12}, A_{13} = 0$, the system of equations simplifies to

$$\begin{aligned} L &= \{A_{11} \cdot A_{21} = 0, A_{11}A_{31} = 0\} \\ &= \{\lambda_{13}\lambda_{32} - \tilde{\lambda}_{12} = 0, -\tilde{\lambda}_{13} + \lambda_{13} = 0\}. \end{aligned}$$

We see that L is (uniquely) solved by $\tilde{\lambda}_{12} = \lambda_{13}\lambda_{32}$ and $\tilde{\lambda}_{13} = \lambda_{13}$ for all $\Lambda \in \mathbb{R}^{G_D}$.

For the other direction, consider $G := G_2$ and $\tilde{G} := G_3$. To check whether $\mathcal{M}(G) \subseteq \mathcal{M}(\tilde{G})$ now, we calculate

$$A = \begin{pmatrix} 1 & 0 & 0 \\ \lambda_{12} - \tilde{\lambda}_{32}\lambda_{13} & 1 & -\tilde{\lambda}_{32} \\ -\tilde{\lambda}_{13} + \lambda_{13} & 0 & 1 \end{pmatrix}.$$

The system of equations simplifies to

$$\begin{aligned} L &= \{A_{11} \cdot A_{21} = 0, A_{11}A_{31} = 0\} \\ &= \{\lambda_{12} - \tilde{\lambda}_{32}\lambda_{13} = 0, -\tilde{\lambda}_{13} + \lambda_{13} = 0\}. \end{aligned}$$

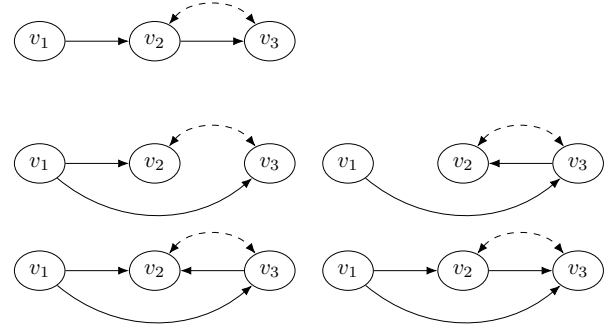


Figure 2: The equivalence class up to closure of the IV graph. Upper left: G_1 (IV Graph), middle left: G_2 , middle right: G_3 , bottom left: G_4 , bottom right: G_5 .

If $\lambda_{13} \neq 0$, L has solution $\tilde{\lambda}_{13} = \lambda_{13}$ and $\tilde{\lambda}_{32} = \frac{\lambda_{12}}{\lambda_{13}}$. However, if $\lambda_{13} = 0$ and $\lambda_{12} \neq 0$, then L has no solution. Due to these exceptional cases, $\mathcal{M}(G_2) \not\subseteq \mathcal{M}(G_3)$.

The fully symbolic Algorithm 1 requires careful inspection of how solvability of L depends on the choice of Λ . However, it is also informative to take a computationally more tractable approach and check whether L has a solution for one (or possibly several) randomly chosen $\Lambda \in \mathbb{R}^{G_D}$, where the entries λ_{ij} follow a continuous distribution, such as being independently drawn from a uniform distribution on $(-C, C)$ for $C > 0$. This allows one to check whether the equation system in Theorem 3.1 has a solution for all but a Lebesgue null set of choices of Λ . We obtain a check of the following relaxed notion of model equivalence.

Definition 3.1 (Equivalence up to closure). *For two ADMGs G and $\tilde{G}$, we say that model $\mathcal{M}(G)$ is contained in the closure of model $\mathcal{M}(\tilde{G})$ if there is a null set $N \subseteq \mathbb{R}^{G_D}$ such that the joint distribution of every random vector $(I - \Lambda)^{-T}\varepsilon$ with $\Lambda \in \mathbb{R}^{G_D} \setminus N$ and $\varepsilon \in \mathcal{M}(G_B)$ is also the joint distribution of a random vector in $\mathcal{M}(\tilde{G})$. We say that G and $\tilde{G}$ are model equivalent up to closure if $\mathcal{M}(G)$ and $\mathcal{M}(\tilde{G})$ are contained in one another up to closure.*

In the sequel, when considering model equivalence, we will always understand it as equivalence up to closure. We will emphasize this fact only in highlighted theorems.

Example 3.3. Figure 2 shows all graphs that are model equivalent (up to closure) to the IV graph.

A practical implementation of the randomized version of Algorithm 1 in the computer algebra system ‘Macaulay2’ [Grayson and Stillman, 1993] will be discussed in Section 4.

3.2 EQUIVALENCE AFTER REMOVAL OF A DIRECTED EDGE

A graphical characterization of model equivalence remains still unclear. In the classical linear Gaussian setting, the

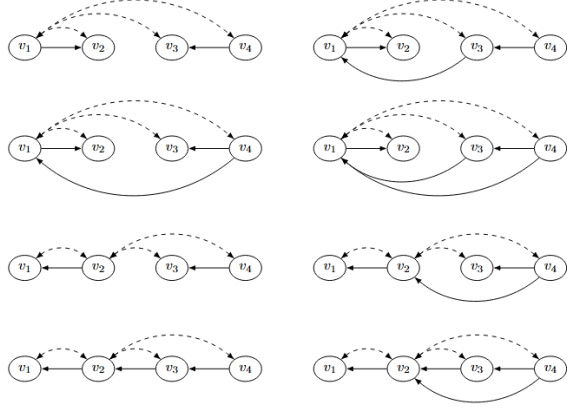


Figure 3: An equivalence class of ADMGs with four vertices. From upper left, row by row: $G_1, G_2, \dots, G_8$.

Markov equivalence class for DAGs is fully characterized by *d-separation*. For MAGs (Maximal Ancestral Graphs) and ADMGs, there also exist algorithms to check Markov equivalence [Richardson and Spirtes, 2002, Wienöbst et al., 2022]. However, for our more refined equivalence classes, checking d-separation relations is clearly not sufficient. We thus consider a simpler problem and investigate equivalence after a minimal change, namely, removing a directed edge.

Figure 3 depicts an equivalence class (up to closure) of ADMGs with four vertices. All ADMGs in this class can be derived from G_4 or G_8 by removing directed edges.

Definition 3.2 (Removable ancestors). *The set of removable ancestors of a node $v \in V$ is defined as*

$$\begin{aligned} R_v &:= \{u \in an(v) : Sib(u) \setminus Sib(v) \neq \emptyset\} \\ &= Sib(V \setminus Sib(v)) \cap an(v). \end{aligned}$$

Definition 3.3 (v-rank [Tramontano et al., 2024a]). *Let $v \in V$, and let $Q \subseteq (pa(v) \cup \{v\})$. The v-rank of Q is*

$$\begin{aligned} r_Q^v &:= \max \{k \in \mathbb{N}_0 : I \subseteq R_v, P \subseteq Q, \\ &\quad |I| = |P| = k, \tilde{P}(I, P) \neq \emptyset\}. \end{aligned}$$

Here, $\tilde{P}(I, P)$ denotes the set of non-intersecting systems of paths between I and P .

Theorem 3.2. *Let $G = (V, E_{\rightarrow}, E_{\leftrightarrow})$ be an ADMG containing edge $e = u_0 \rightarrow v_0 \in E_{\rightarrow}$. Let $G^e := (V, E_{\rightarrow} \setminus \{e\}, E_{\leftrightarrow})$. Then G and G^e are model equivalent up to closure if and only if*

$$r_{pa(v_0) \setminus u_0}^{v_0} = r_{pa(v_0)}^{v_0}. \quad (3.1)$$

Example 3.4. Consider G_1 and G_5 from Figure 2. Let $G := G_5$ and $G^e = G_1$. Let $e = v_1 \rightarrow v_3$ be the removed edge. By definition, $R_{v_3} = Sib(V \setminus Sib(v_3)) \cap an(v) = \{v_1\}$.

Further, we have

$$\begin{aligned} r_{pa(v_3) \setminus v_1}^{v_3} &= r_{v_2}^{v_3} = \max \{k \in \mathbb{N}_0 : I \subseteq \{v_1\}, P \subseteq \{v_2\}, \\ &\quad |I| = |P| = k, \tilde{P}(I, P) \neq \emptyset\} = 1. \end{aligned}$$

Similarly, $r_{pa(v_3)}^{v_3} = 1$. So we conclude that removing $v_1 \rightarrow v_3$ from G keeps it in the same equivalence class. Using the same approach, we can also verify that $\mathcal{M}(G_5) = \mathcal{M}(G_2)$ by removing $v_2 \rightarrow v_3$, $\mathcal{M}(G_4) = \mathcal{M}(G_2)$ by removing $v_3 \rightarrow v_2$, and $\mathcal{M}(G_4) = \mathcal{M}(G_3)$ by removing $v_1 \rightarrow v_2$. As illustrated in Figure 4, the whole equivalence class of the IV graph in Figure 2 can be derived from any ADMG in this class by sequentially adding or removing edges.

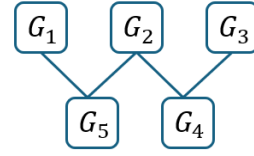


Figure 4: The relation of $G_1 \dots, G_5$ from Figure 2. An undirected edge between two ADMGs represents that one could be derived from another by adding or removing a directed edge.

Theorem 3.2 provides a graphical check if a directed edge is removable when considering model equivalence. While verifying Equation (3.1) directly could be computationally challenging, Tramontano et al. [2024a] proposed algorithms that can verify such a condition in polynomial time in the size of the graph via a maximum flow reformulation. In fact, (3.1) is related to the concept of generic identifiability (Tramontano et al. [2024a, §2]). Essentially, Theorem 3.2 states that a directed edge $u \rightarrow v$ can be removed without changing the equivalence class if and only if the parameter λ_{uv} is not generically identifiable (see Theorem A.1).

3.3 A NECESSARY CONDITION FOR THE BIDIRECTED PART

Intuitively, by the nonparametric nature of our confounding model, two ADMGs G and $\tilde{G}$ should only be model equivalent if they have the same number of bidirected edges. This necessary condition on the bidirected part can be generalized by considering higher moments, which is most conveniently done using *cumulant tensors* [cf. Robeva and Seby, 2021].

Definition 3.4 (Cumulant tensor). *The k -th cumulant tensor of a random vector $\varepsilon = (\varepsilon_1, \dots, \varepsilon_p)$ is the k -way tensor in $(\mathbb{R}^p)^k$ with*

$$\begin{aligned} \mathcal{C}^{(k)}(\varepsilon)_{i_1, \dots, i_k} &:= \sum_{(A_1, \dots, A_L)} (-1)^{L-1} (L-1)! \\ &\quad \cdot \mathbb{E} \left[\prod_{j \in A_1} \varepsilon_j \right] \cdots \mathbb{E} \left[\prod_{j \in A_L} \varepsilon_j \right], \end{aligned}$$

where the sum is taken over all partitions $(A_1, \dots, A_L)$ of the multiset $\{i_1, \dots, i_k\}$ with arbitrary $L \in \{1, \dots, k\}$.

Since we always center ε , it holds that $\mathcal{C}^{(1)}(\varepsilon) = \mathbb{E}[\varepsilon] = 0$. For $k = 2$, $\mathcal{C}^{(2)}(\varepsilon)$ is the covariance matrix. Moreover, $\mathcal{C}^{(3)}(\varepsilon)_{ijk} = \mathbb{E}[\varepsilon_i \varepsilon_j \varepsilon_k]$ and

$$\begin{aligned} \mathcal{C}^{(4)}(\varepsilon)_{ijkl} &= \mathbb{E}[\varepsilon_i \varepsilon_j \varepsilon_k \varepsilon_l] - \mathbb{E}[\varepsilon_i \varepsilon_j] \mathbb{E}[\varepsilon_k \varepsilon_l] \\ &\quad - \mathbb{E}[\varepsilon_i \varepsilon_k] \mathbb{E}[\varepsilon_j \varepsilon_l] - \mathbb{E}[\varepsilon_i \varepsilon_l] \mathbb{E}[\varepsilon_j \varepsilon_k] \end{aligned}$$

after centering. Cumulant tensors are symmetric, i.e.,

$$\mathcal{C}^{(k)}(\varepsilon)_{i_1, \dots, i_k} = \mathcal{C}^{(k)}(\varepsilon)_{\sigma(i_1), \dots, \sigma(i_k)} \quad \forall \sigma \in S_k,$$

where S_k is the symmetric group on $\{1, \dots, k\}$. We write $\text{Sym}_k(p)$ for the subspace of symmetric tensors in $(\mathbb{R}^p)^k$. We now introduce the cumulant tensor space of G_B and G .

Definition 3.5 (Cumulant tensor space). *Let*

$$\begin{aligned} \mathcal{T}^{(k)}(G_B) &:= \{\mathcal{C}^{(k)} \in \text{Sym}_k(\mathbb{R}^p) : \mathcal{C}^{(k)}_{i_1, \dots, i_k} = 0 \\ &\quad \text{if } \{i_1, \dots, i_k\} \text{ is not connected in } G_B\}, \end{aligned}$$

and let

$$\mathcal{T}^{(k)}(G) := \{\mathcal{C}^{(k)} \bullet_k B_\Lambda \mid \mathcal{C}^{(k)} \in \mathcal{T}^{(k)}(G_B), \Lambda \in \mathbb{R}^{G_D}\},$$

where $\bullet_k$ denotes the Tucker product (see Lemma A.4).

Definition 3.6. *Let $G = (V, E_{\rightarrow}, E_{\leftrightarrow})$ be an ADMG, and let G_B be its bidirected part. For all $i \in \mathbb{N}_{>0}$, define*

$$n_i(G_B) := |\{S \subseteq V : S \text{ connected in } G_B, |S| = i\}|,$$

i.e., the number of connected size- i subsets of V with respect to G_B .

Note that $n_1(G_B)$ is simply the number of vertices, since each vertex is connected to itself via the trivial path. Further, $n_2(G_B)$ is the number of bidirected edges. In general, $n_i(G_B)$ is bounded by the binomial coefficient $\binom{p}{i}$, where $p = |V|$. By definition, $n_i(G_B) = 0$ for $i > p$.

As an example, for the ADMGs in Figure 5, $n_1(G_B) = n_1(\tilde{G}_B) = 5$, $n_2(G_B) = n_2(\tilde{G}_B) = 3$, $n_3(G_B) = n_3(\tilde{G}_B) = 1$ and $n_i(G_B) = n_i(\tilde{G}_B) = 0$ for all $i \geq 4$.

The following lemma shows that the dimension of $\mathcal{T}^{(k)}(G_B)$ is solely determined by $n_i(G_B)$ with $1 \leq i \leq \min(p, k)$, $i \in \mathbb{N}_{>0}$.

Lemma 3.3. *Let G_B be the bidirected part of an ADMG G . Then*

$$\dim(\mathcal{T}^{(k)}(G_B)) = \sum_{i=1}^{\min\{p, k\}} n_i(G_B) \cdot \binom{k-1}{i-1}.$$

Example 3.5. Consider $\dim(\mathcal{T}^{(3)}(G_B))$ in Figure 5. The twelve non-zero cumulant tensors are $\mathcal{C}_{1,1,1}, \dots, \mathcal{C}_{5,5,5}$,

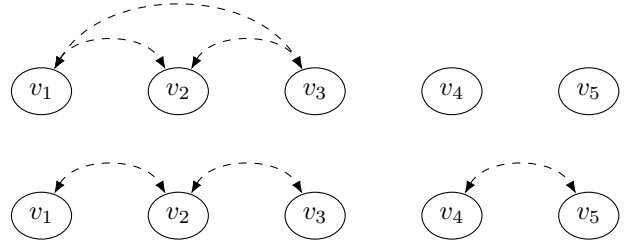


Figure 5: Bidirected parts of two ADMGs: G_B (above) and $\tilde{G}_B$ (below).

$\mathcal{C}_{1,1,2}, \mathcal{C}_{1,2,2}, \mathcal{C}_{1,1,3}, \mathcal{C}_{1,3,3}, \mathcal{C}_{2,2,3}, \mathcal{C}_{2,3,3}$ and $\mathcal{C}_{1,2,3}$, by ignoring all permutations. This result corresponds to the formula

$$\begin{aligned} \dim(\mathcal{T}^{(3)}(G_B)) &= \sum_{i=1}^{\min\{5,3\}} n_i(G_B) \cdot \binom{3-1}{i-1} \\ &= 5 \cdot \binom{2}{0} + 3 \cdot \binom{2}{1} + 1 \cdot \binom{2}{2} = 12. \end{aligned}$$

Essentially, when two ADMGs are model equivalent, their corresponding cumulant tensor spaces should be of the same dimension for all $k \in \mathbb{N}_{>0}$. Our model parametrization restricts the dimensional difference of their corresponding bidirected cumulant tensor spaces. Applying the dimension formula above yields our next theorem, which provides a necessary graphical condition on the bidirected parts of two ADMGs for model equivalence.

Theorem 3.4. *If two ADMGs G and $\tilde{G}$ are model equivalent (up to closure), then $n_i(G_B) = n_i(\tilde{G}_B)$ for all $i \in \mathbb{N}_{>0}$.*

By Theorem 3.4, two ADMGs with bidirected parts as in Figure 6 cannot be model equivalent because $n_3(G_B) = 4 \neq 3 = n_3(\tilde{G}_B)$. By applying both Theorem 3.1 and Theorem 3.4, one can show that the four cycle $G = G_B$ as in Figure 6 is identifiable (see Example A.1). Note that the condition $n_i(G_B) = n_i(\tilde{G}_B)$ for all $i \in \mathbb{N}^+$ in Theorem 3.4 does not imply that the two bidirected parts G_B and $\tilde{G}_B$ have the same structure (also not up to symmetry); Figure 5 shows a counterexample.

Consider a disjoint decomposition of G_B such that $V = \cup_{i=1}^m D_i$, where D_i is connected in G_B . If D_i is not further contained in a larger bidirected-connected subset, then D_i is maximal and often called a *district* in G . Let

$$\mathcal{D}(G) := \{D : D \text{ is a district in } G\}$$

be the set of districts in G . Let

$$D_{\max} := \{D \in \mathcal{D}(G) : |D| \geq |D_i| \quad \forall D_i \in \mathcal{D}(G)\}$$

be the maximal district of G . A straightforward application of Theorem 3.4 yields:

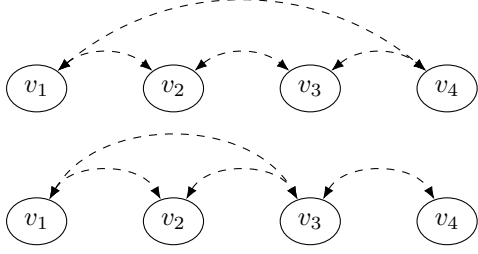


Figure 6: Non-equivalence: G_B (above) and $\tilde{G}_B$ (below).

Corollary 3.5. *If G and $\tilde{G}$ are model equivalent ADMGs, then:*

- a) $|E_{\leftrightarrow}| = |\tilde{E}_{\leftrightarrow}|$,
- b) $|D_{\max}| = |\tilde{D}_{\max}|$,
- c) $\dim(\mathcal{T}^{(k)}(G_B)) = \dim(\mathcal{T}^{(k)}(\tilde{G}_B)) \quad \forall k \in \mathbb{N}_{>0}$.

Theorem 3.4 provides a necessary condition for checking model equivalence. We aim to investigate its computational complexity. Let G and $\tilde{G}$ be two arbitrary ADMGs with p nodes. Ideally, we want to derive an efficient algorithm for the non-equivalence filter

$$f(G, \tilde{G}) \mapsto \begin{cases} \text{Continue} & \text{if } n_i(G_B) = n_i(\tilde{G}_B) \quad \forall i \in [p], \\ \text{False} & \text{otherwise.} \end{cases}$$

To compute $n_i(G_B)$, we can naively go through all $\binom{p}{i}$ subsets and decide in each case whether it is connected or not. Therefore, a naive computation of f by calculating all $n_i(G_B)$ and $n_i(\tilde{G}_B)$ for $i \in [p]$ would require checking the connectedness of

$$\mathcal{O}\left(\sum_{i=2}^p \binom{p}{i}\right) \sim \mathcal{O}(2^p)$$

subsets. We may check whether a size i subset of nodes is connected in complexity $\mathcal{O}(i^2)$.

We are more interested in the case $f = \text{False}$, where we can immediately conclude non-equivalence. A possible application of Theorem 3.4 is to add fast rejection criteria to Algorithm 1. A straightforward rejection filter is to check the number of bidirected edges, i.e., $n_2(G_B)$. However, computing $n_i(G_B)$ for moderate i can be expensive. It is worth noting, however, that a largest connected component can be found by a DFS/BFS traversal in $\mathcal{O}(p + |E_{\leftrightarrow}|)$ [Tarjan, 1972], which suggests checking $|D_{\max}| = |\tilde{D}_{\max}|$ as in Corollary 3.5. Another observation is that an algorithm to compute $n_i(G)$ is output sensitive, i.e., the time needed depends on the value of $n_i(G)$ itself. Komusiewicz and Sommer [2021] proposed an algorithm with solution delay $\mathcal{O}(i^2 \Delta)$ where Δ is the maximum degree in the input graph. Therefore, if G_B is sparse, one would expect a fast return of f .

4 COMPUTATIONAL EXPERIMENTS

‘Macaulay2’ [Grayson and Stillman, 1993] is a computer algebra system that is commonly used to support research in graphical models [Amendola et al., 2013]. Using randomization and results from Corollary 3.5, we implemented a revised version of Algorithm 1 in ‘Macaulay2’, resulting in Algorithm 2 (Code available under https://anonymous.4open.science/r/Model_Equivalence-6E66).

Algorithm 2 Check if $\mathcal{M}(G) = \mathcal{M}(\tilde{G})$

```

if  $|E_{\leftrightarrow}| \neq |\tilde{E}_{\leftrightarrow}|$  then
    return  $\mathcal{M}(G) \neq \mathcal{M}(\tilde{G})$ 
end if
Compute connected components of  $G_B$  and  $\tilde{G}_B$ ;
if  $D_{\max} \neq \tilde{D}_{\max}$  then
    return  $\mathcal{M}(G) \neq \mathcal{M}(\tilde{G})$ 
end if
 $A \leftarrow (I - \tilde{\Lambda})^T (I - \Lambda)^{-T}$ ;
 $\tilde{A} \leftarrow (I - \Lambda)^T (I - \tilde{\Lambda})^{-T}$ ;
 $L, \tilde{L} \leftarrow \emptyset$ ;
for each  $u \leftrightarrow v \notin \tilde{G}$  with  $v < u$  do
    for each  $w_1 = w_2$  and  $w_1 \leftrightarrow w_2 \in G$  do
        if  $A_{uw_1} \neq 0$  and  $A_{vw_2} \neq 0$  then
             $L \leftarrow L \cup \{A_{uw_1} \cdot A_{vw_2} = 0\}$ ;
        end if
    end for
end for
for each  $u \leftrightarrow v \notin G$  with  $v < u$  do
    for each  $w_1 = w_2$  and  $w_1 \leftrightarrow w_2 \in \tilde{G}$  do
        if  $\tilde{A}_{uw_1} \neq 0$  and  $\tilde{A}_{vw_2} \neq 0$  then
             $\tilde{L} \leftarrow \tilde{L} \cup \{\tilde{A}_{uw_1} \cdot \tilde{A}_{vw_2} = 0\}$ ;
        end if
    end for
end for
for  $\lambda \in L, \tilde{\lambda} \in \tilde{L}$  do
    substitute  $\lambda, \tilde{\lambda} \stackrel{iid}{\sim} \text{Uniform}(-C, C)$  for a  $C > 0$ ;
end for
if  $L$  has a solution for  $\tilde{\lambda}$  and  $\tilde{L}$  has a solution for  $\lambda$  then
    return  $\mathcal{M}(G) = \mathcal{M}(\tilde{G})$ 
else
    return  $\mathcal{M}(G) \neq \mathcal{M}(\tilde{G})$ 
end if

```

The revised algorithm proceeds as follows: When generating the equation systems L and $\tilde{L}$, we assume that there exists an ordering on the vertex set. For a system L , instead of adding all $u \leftrightarrow v \notin \tilde{G}$ to the system, we realize the symmetry and consider only (u, v) as a candidate if $u > v$. It is also worth noting that $A_{ij} = 0$ if $i \neq \text{de}(j)$. If A_{uw_1} or A_{vw_2} equals to zero, this equation would be omitted. Finally, to solve L and $\tilde{L}$, we substitute λ and $\tilde{\lambda}$ with randomly generated numbers; for computational algebra we present this as rational numbers.

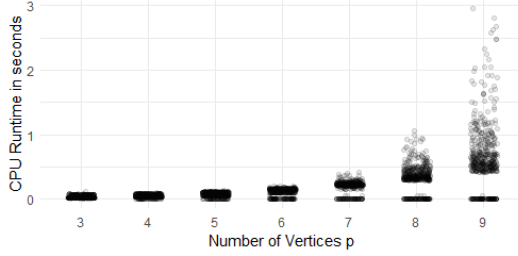


Figure 7: Raw data of CPU time using Algorithm 2, for $p \in \{3, 4, \dots, 9\}$.

To evaluate the performance of Algorithm 2, we run random experiments for different numbers of vertices ($p = 3, 4, \dots, 9$) and record for each case the *CPU processing time* in seconds. For each fixed p , we repeat the process of uniformly randomly sampling two ADMGs with the same number of bidirected edges for 500 times and record the results as in Figure 7, where each data point represents an experiment. Figure 9 in the Appendix shows the computation time on a log scale and shows an exponential increase in average CPU time.

The distribution of CPU time is bimodal after adding rejection filters (Figure 8). Table 1 is obtained by fixing $p = |V| = 7$ and varying the number of bidirected edges. For each different $|E_{\leftrightarrow}|$, we sample 200 pairs of ADMGs with the given $|E_{\leftrightarrow}|$ and run Algorithm 2 on these pairs. We record the mean, median, minimum, and maximum CPU processing time of these samples. Since the size of the equation system $|L|$ in Algorithm 2 decreases as the number of bidirected edges increases, we observe a more extreme maximal CPU time when $|E_{\leftrightarrow}|$ is small. At the same time, when $|E_{\leftrightarrow}|$ is small, the fast rejection via maximal connected component is more likely to happen as we observe also a more extreme minimal CPU time. Using our code, we separated all $25 \times 2^{\binom{3}{2}} = 200$ ADMGs with 3 vertices into 74 equivalence classes (see Appendix C). 25 classes among them consist of a unique ADMG without bidirected edges. These are the 25 DAGs with three vertices (which are identifiable for linear non-Gaussian models). Another trivial class is the one that consists of all 25 ADMGs with fully connected bidirected parts.

5 CONCLUSIONS

In this work, we studied equivalence of ADMG models in a linear non-Gaussian framework with general latent confounding. We showed how to characterize model equivalence, possibly up to closure, using algebraic methods. Moreover, we characterized removability of directed edges and developed a necessary condition for model equivalence based on the connectedness of the bidirected parts. Although this necessary condition is straightforward to verify graphi-

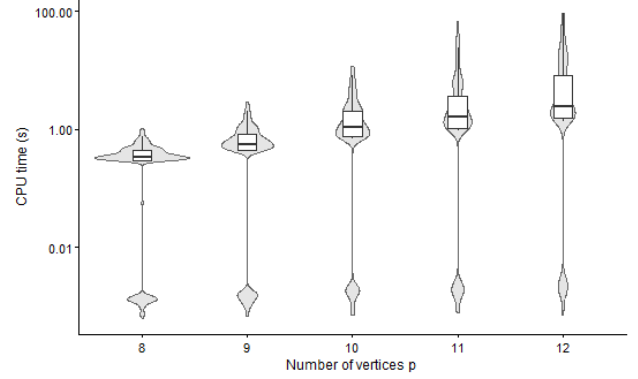


Figure 8: Distribution of the CPU time (with the y-axis in log scale) for moderate size of graphs, i.e., $p \in \{8, 9, 10, 11, 12\}$, using Algorithm 2.

$ E_{\leftrightarrow} $	Mean	Median	Min	Max
3	0.122	0.0473	0.000692	0.935
4	0.0890	0.00119	0.000752	0.540
5	0.0971	0.00135	0.000769	0.402
6	0.0950	0.00135	0.000874	0.444
7	0.114	0.00139	0.000975	0.350
8	0.156	0.222	0.000982	0.401
9	0.184	0.230	0.00109	0.360
10	0.216	0.232	0.00102	0.426
11	0.233	0.232	0.00106	0.482
12	0.241	0.235	0.00115	0.433
15	0.244	0.237	0.164	0.411
18	0.238	0.228	0.162	0.441

Table 1: Execution time in seconds with different numbers of bidirected edges, where we fix $|V| = 7$.

cally, its practical implementation is nontrivial and requires careful handling because of the high algorithmic complexity.

Interesting topics for future work include potential sharpening of graphical conditions for model equivalence of ADMGs. Moreover, it would be of interest to generalize our results to cyclic setups such as considered, e.g., in Drton et al. [2025] or Dai et al. [2024].

References

- Carlos Améndola, Luis David García Puente, Roser Homs Pons, Olga Kuznetsova, Harshit J Motwani, Sonja Petrovic, Mike Stillman, and Seth Sullivant. Graphical Models. *The Journal of Software for Algebra and Geometry*, 5, 2013.
- Carlos Améndola, Philipp Dettling, Mathias Drton, Federica Onori, and Jun Wu. Structure learning for cyclic linear causal models. In *Proceedings of the Thirty-Sixth Conference on Uncertainty in Artificial Intelligence, UAI 2020, virtual online, August 3-6, 2020*, volume 124 of

- Proceedings of Machine Learning Research*. AUAI Press, 2020.
- Rina Foygel Barber, Mathias Drton, Nils Sturman, and Luca Weihs. Half-trek criterion for identifiability of latent variable models. *Ann. Statist.*, 50, 2022.
- Haoyue Dai, Ignavier Ng, Yujia Zheng, Zhengqing Gao, and Kun Zhang. Local causal discovery with linear non-Gaussian cyclic models. In Sanjoy Dasgupta, Stephan Mandt, and Yingzhen Li, editors, *Proceedings of The 27th International Conference on Artificial Intelligence and Statistics*, volume 238 of *Proceedings of Machine Learning Research*. PMLR, 2024.
- G. Darmon. Analyse générale des liaisons stochastiques. Etude particulière de l’analyse factorielle linéaire. *Rev. Inst. Internat. Statist.*, 21, 1953.
- Xinshuai Dong, Ignavier Ng, Biwei Huang, Yuewen Sun, Songyao Jin, Roberto Legaspi, Peter Spirtes, and Kun Zhang. On the parameter identifiability of partially observed linear causal models. In *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*, 2024.
- Mathias Drton. Algebraic problems in structural equation modeling. In *The 50th anniversary of Gröbner bases*, volume 77 of *Adv. Stud. Pure Math.* Math. Soc. Japan, Tokyo, 2018.
- Mathias Drton and Thomas S. Richardson. Binary models for marginal independence. *J. R. Stat. Soc. Ser. B Stat. Methodol.*, 70(2), 2008.
- Mathias Drton, Elina Robeva, and Luca Weihs. Nested covariance determinants and restricted trek separation in Gaussian graphical models. *Bernoulli*, 26, 2020.
- Mathias Drton, Marina Garrote-López, Niko Nikov, Elina Robeva, and Y. Samuel Wang. Causal discovery for linear non-Gaussian models with disjoint cycles. In *Conference on Uncertainty in Artificial Intelligence, Rio Othon Palace, Rio de Janeiro, Brazil, 21-25 July 2025*, volume 286 of *Proceedings of Machine Learning Research*, 2025.
- Daniel R. Grayson and Michael E. Stillman. Macaulay2, a software system for research in algebraic geometry, 1993.
- C. Jutten and P. Comon. *Handbook of Blind Source Separation*. Academic Press, Oxford, 2010.
- Christian Komusiewicz and Frank Sommer. Enumerating connected induced subgraphs: Improved delay and experimental comparison. *Discrete Applied Mathematics*, 303, 2021.
- Judea Pearl. Linear models: A useful “microscope” for causal analysis. *Journal of Causal Inference*, 1, 2013.
- Thomas Richardson. Markov properties for acyclic directed mixed graphs. *Scand. J. Statist.*, 30, 2003.
- Thomas Richardson and Peter Spirtes. Ancestral graph Markov models. *Ann. Statist.*, 30, 2002.
- Thomas S. Richardson, Robin J. Evans, James M. Robins, and Ilya Shpitser. Nested Markov properties for acyclic directed mixed graphs. *Ann. Statist.*, 51, 2023.
- Elina Robeva and Jean-Baptiste Seby. Multi-trek separation in linear structural equation models. *SIAM J. Appl. Algebra Geom.*, 5, 2021.
- Saber Salehkaleybar, AmirEmad Ghassami, Negar Kiyavash, and Kun Zhang. Learning linear non-gaussian causal models in the presence of latent variables. *Journal of Machine Learning Research*, 21, 2020.
- Shohei Shimizu. *Statistical causal discovery: LiNGAM approach*. Springer Japan, Tokyo, 2022. JSS Research Series in Statistics.
- Ilya Shpitser, Robin J. Evans, and Thomas S. Richardson. Acyclic linear sems obey the nested Markov property. In *Proceedings of the Thirty-Fourth Conference on Uncertainty in Artificial Intelligence, UAI 2018, Monterey, California, USA, August 6-10, 2018*, 2018.
- V. P. Skitovich. Linear forms of independent random variables and the normal distribution law. *Izv. Akad. Nauk SSSR Ser. Mat.*, 18, 1954.
- Peter Spirtes and Kun Zhang. Search for causal models. In S. Lauritzen M. Maathuis, M. Drton and M. Wainwright, editors, *Handbook of graphical models*, Chapman & Hall/CRC Handb. Mod. Stat. Methods. CRC Press, Boca Raton, FL, 2019.
- Seth Sullivant, Kelli Talaska, and Jan Draisma. Trek separation for Gaussian graphical models. *Ann. Statist.*, 38, 2010.
- Robert Tarjan. Depth-first search and linear graph algorithms. *SIAM Journal on Computing*, 1, 1972.
- Daniele Tramontano, Mathias Drton, and Jalal Etesami. Parameter identification in linear non-Gaussian causal models under general confounding, 2024a. To appear, <https://imstat.org/journals-and-publications/annals-of-statistics/annals-of-statistics-future-papers>.
- Daniele Tramontano, Yaroslav Kivva, Saber Salehkaleybar, Mathias Drton, and Negar Kiyavash. Causal effect identification in LiNGAM models with latent confounders. In

Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024, volume 235 of *Proceedings of Machine Learning Research*, 2024b.

Thijs van Ommen. Efficiently deciding algebraic equivalence of bow-free acyclic path diagrams. In *Uncertainty in Artificial Intelligence, 15-19 July 2024, Universitat Pompeu Fabra, Barcelona, Spain*, volume 244 of *Proceedings of Machine Learning Research*, 2024.

Thomas Verma and Judea Pearl. Equivalence and synthesis of causal models. In *UAI '90: Proceedings of the Sixth Annual Conference on Uncertainty in Artificial Intelligence, MIT, Cambridge, MA, USA, July 27-29, 1990*, 1990.

Y. Samuel Wang and Mathias Drton. Causal discovery with unobserved confounding and non-Gaussian data. *J. Mach. Learn. Res.*, 24, 2023.

Marcel Wienöbst, Max Bannach, and Maciej Liśkiewicz. A new constructive criterion for Markov equivalence of MAGs. In *Proceedings of the Thirty-Eighth Conference on Uncertainty in Artificial Intelligence*, volume 180, 2022.

Qingyuan Zhao. On statistical and causal models associated with acyclic directed mixed graphs. *arXiv:2501.03048*, 2025.

Model Equivalence for Linear Non-Gaussian Causal Models Under General Confounding (Supplementary Material)

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A PROOFS FOR THE MAIN PAPER

Proof for Theorem 3.1.

Proof. By definition, $\mathcal{M}(G) \subseteq \mathcal{M}(\tilde{G})$ if and only if for every $X = (I - \Lambda)^{-T} \varepsilon \in \mathcal{M}(G)$, we have $X \stackrel{d}{=} (I - \tilde{\Lambda})^{-T} \tilde{\varepsilon}$ for some $\tilde{\Lambda} \in \mathbb{R}^{\tilde{G}_D}$ and $\tilde{\varepsilon} \in \mathcal{M}(\tilde{G}_B)$. This implies

$$\tilde{\varepsilon} \stackrel{d}{=} (I - \tilde{\Lambda})^T (I - \Lambda)^{-T} \varepsilon = \underbrace{(I - \tilde{\Lambda})^T B_\Lambda}_A \varepsilon. \quad (\text{A.1})$$

In particular, Equation (A.1) has to hold for $\varepsilon \in \mathcal{M}(G_B)$ satisfying Equation (2.1). By applying Equation (2.1) to every pair $u \leftrightarrow v \notin \tilde{G}$, we conclude that the condition is necessary for the model inclusion. In order to prove its sufficiency, we must demonstrate that $\tilde{\varepsilon}$ satisfies the connected set Markov property with respect to G .

For every $v \in V$, let $D(v) = \{u \in V : a_{vu} \neq 0\}$, and $D(C) := \cup_{v \in C} D(v)$ for $C \subseteq V$. We need to show that $\tilde{\varepsilon}_C \perp\!\!\!\perp \tilde{\varepsilon}_{V \setminus \text{Sib}(C)}$ for every $C \subseteq V$ that is connected in $\tilde{G}_B$. If $D(V \setminus \text{Sib}(C)) \subseteq V \setminus \text{Sib}(D(C))$ in G , then the result follows from the connected set Markov property of ε with respect to G . Assume, for contradiction, that $D(V \setminus \text{Sib}(C)) \cap \text{Sib}(D(C))$, indicating that there exist $w_1 \in D(V \setminus \text{Sib}(C))$, $w_2 \in D(C)$, $u \in V \setminus \text{Sib}(C)$, and $v \in C$ such that $w_1 \leftrightarrow w_2$ in G or $w_1 = w_2$, and $a_{uw_1} \cdot a_{vw_2} \neq 0$, which is a contradiction since $u \leftrightarrow v \notin \tilde{G}$. $\square$

Let G be an ADMG. Let $X \in \mathcal{M}(G)$ be a distribution parametrized by Φ_G . Let the *fiber* of X be the set

$$\Phi_G^{-1}(X) = \{(\Lambda, \varepsilon) \in \mathbb{R}^{G_D} \times \mathcal{M}(G_B) : \Phi_G(\Lambda, \varepsilon) \stackrel{d}{=} X\}.$$

Let $\mathbf{P}_{\mathbb{R}^{G_D}}(\Phi_G^{-1}(X))$ denote the projection of this set onto $\mathbb{R}^{G_D}$.

Definition A.1 (generic identifiability). *A parameter given by a function $f(\Lambda)$ is generically identifiable if any generic choice of (Λ, ε) yields a random vector $X = (I - \Lambda)^{-T} \varepsilon$ for which it holds that $f(\tilde{\Lambda}) = f(\Lambda)$ for all $\tilde{\Lambda} \in \mathbf{P}_{\mathbb{R}^{G_D}}(\Phi_G^{-1}(X))$.*

Requiring genericity of $\Lambda \in \mathbb{R}^{G_D}$ allows us to exclude a fixed Lebesgue null set of $\mathbb{R}^{G_D}$. Note that excluding null sets of a finite-dimensional space is a standard approach in related literature ([Drton, 2018, §8]; [Tramontano et al., 2024a, §2]).

Theorem A.1 (compare Tramontano et al. [2024a, §3, Theorem 3.3]). *Let $u \in pa(v)$. The parameter λ_{uv} is generically identifiable if and only if*

$$r_{pa(v) \setminus u}^v = r_{pa(v)}^v - 1,$$

where r_Q^v denotes the v-rank of Q as in Definition 3.3.

Lemma A.2 (compare Tramontano et al. [2024a, §3, Lemma 3.1]). *Let G be an ADMG, and let $X = \Phi_G(\Lambda, \varepsilon)$ for $(\Lambda, \varepsilon) \in \mathbb{R}^{G_D} \times \mathcal{M}(G_B)$. Suppose X can be generated using another pair $(\tilde{\Lambda}, \tilde{\varepsilon}) \in \Phi_G^{-1}(X)$. Then we have $\tilde{\varepsilon} \stackrel{d}{=} A\varepsilon$ where*

$A = (I - \tilde{\Lambda})^T B_\Lambda$, and its entries can be specified as a function of $\tilde{\Lambda}$ and B_Λ through the ancestral relations among the nodes of G :

$$\begin{aligned} a_{vu} &= b_{vu} - \sum_{w \in V} \tilde{\lambda}_{wv} b_{wu} \\ &= b_{vu} - \sum_{w \in \text{pa}(v) \cap \text{de}(u)} \tilde{\lambda}_{wv} b_{wu}. \end{aligned}$$

In particular, we have $a_{vu} = 0$ if $v \notin \text{de}(u)$, and $a_{uu} = 1$ for every $u \in V$.

Lemma A.3 (Tramontano et al. [2024a, §3, Lemma 3.2]). *Let $X = \Phi_G(\Lambda, \varepsilon)$ for a generic choice of parameters $(\Lambda, \varepsilon) \in \mathbb{R}^{G_D} \times \mathcal{M}(G_B)$. The matrix $\tilde{\Lambda} \in \mathbb{R}^{G_D}$ belongs to $\mathbf{P}_{\mathbb{R}^G}(\Phi_G^{-1}(X))$ if and only if it is a solution to the following linear system of equations:*

$$[(B_\Lambda)_{\text{pa}(v), R_v}]^T \cdot \tilde{\lambda}_{\text{pa}(v), v} = [(B_\Lambda)_{v, R_v}]^T, \quad \forall v \in V.$$

Proof for Theorem 3.2.

Proof. Since G^e is a subgraph of G , we always have $\mathcal{M}(G^e) \subseteq \mathcal{M}(G)$. Hence, by theorem A.1, we want to show that $\mathcal{M}(G^e) \supseteq \mathcal{M}(G)$ if and only if $\lambda_{u_0 v_0}$ is not generically identifiable in G .

Let $X = (I - \Lambda)^{-T} \varepsilon \in \mathcal{M}(G)$. Then $X \in \mathcal{M}(G^e)$ if and only if $X \stackrel{d}{=} (I - \tilde{\Lambda})^{-T} \tilde{\varepsilon}$ for $\tilde{\Lambda} \in \mathbb{R}^{G_D^e}$ and $\tilde{\varepsilon} \in \mathcal{M}(G_B^e)$. This implies

$$\tilde{\varepsilon} \stackrel{d}{=} (I - \tilde{\Lambda})^T (I - \Lambda)^{-T} \varepsilon = \underbrace{(I - \tilde{\Lambda})^T B_\Lambda}_A \varepsilon. \quad (\text{A.2})$$

Using Lemma A.2, we can verify that

$$a_{uv} = b_{uv} - \sum_{w \in \text{pa}(u)^e \cap \text{de}(v)} \tilde{\lambda}_{wu} b_{wv},$$

where $\text{pa}(v)^e$ is the parent set of v in G^e .

Since G and G^e have the same bidirected part, by Lemma A.3, we can conclude that a matrix A as in Equation (A.2) can exist if and only if, for every $v \in V$, the following system has a solution:

$$[(B_\Lambda)_{\text{pa}(v)^e, R_v}]^T \cdot \tilde{\lambda}_{\text{pa}(v)^e, v} = [(B_\Lambda)_{v, R_v}]^T. \quad (\text{A.3})$$

For $v \neq v_0$, we have $\text{pa}(v)^e = \text{pa}(v)$. Hence, the system in Equation (A.3) has always a solution given by $\tilde{\lambda}_{\text{pa}(v)^e, v} = \lambda_{\text{pa}(v), v}$.

For $v = v_0$, we have $\text{pa}(v_0)^e = \text{pa}(v_0) \setminus \{u_0\}$. The system in Equation (A.3) has a solution if and only if $\text{rank}((B_\Lambda)_{\text{pa}(v_0) \setminus \{u_0\}, R_{v_0}}) = \text{rank}((B_\Lambda)_{\text{pa}(v_0) \setminus \{u_0\} \cup \{v_0\}, R_{v_0}})$. Let $\Pi = (\pi_0, \dots, \pi_k)$ be a system of paths without intersection from $I \subseteq R_{v_0}$ to $J \subseteq \text{pa}(v_0)$. If $u_0 \in \Pi$, let π_0 be the path that ends at u_0 , and π_0^* be the path obtained by concatenating π_0 with the edge $u_0 \rightarrow v_0$. By contradiction, $\Pi^* = (\pi_0^*, \dots, \pi_k)$ is a system of non-intersection paths from I to $J \setminus \{u_0\} \cup \{v_0\} \subseteq \text{pa}(v_0) \setminus \{u_0\} \cup \{v_0\}$. This proves that $r_{\text{pa}(v_0) \setminus \{u_0\} \cup \{v_0\}}^{v_0} = r_{\text{pa}(v_0)}^{v_0}$, that implies $\text{rank}((B_\Lambda)_{\text{pa}(v_0) \setminus \{u_0\} \cup \{v_0\}, R_{v_0}}) = \text{rank}((B_\Lambda)_{\text{pa}(v_0), R_{v_0}})$.

From Theorem A.1 we know that $\text{rank}((B_\Lambda)_{\text{pa}(v_0) \setminus \{u_0\}, R_{v_0}}) \geq \text{rank}((B_\Lambda)_{\text{pa}(v_0), R_{v_0}}) - 1$ and that the two equality holds if and only if $\lambda_{u_0 v_0}$ is generically identifiable. Finally, we can write

$$\begin{aligned} \text{rank}((B_\Lambda)_{\text{pa}(v_0), R_{v_0}}) - 1 &\leq \text{rank}((B_\Lambda)_{\text{pa}(v_0) \setminus \{u_0\}, R_{v_0}}) \\ &\leq \text{rank}((B_\Lambda)_{\text{pa}(v_0) \setminus \{u_0\} \cup \{v_0\}, R_{v_0}}) = \text{rank}((B_\Lambda)_{\text{pa}(v_0), R_{v_0}}) \end{aligned}$$

This concludes the proof by noticing that $\text{rank}((B_\Lambda)_{\text{pa}(v_0) \setminus \{u_0\}, R_{v_0}}) = \text{rank}((B_\Lambda)_{\text{pa}(v_0) \setminus \{u_0\} \cup \{v_0\}, R_{v_0}})$ if and only if the first inequality is strict, and that $r_{\text{pa}(v_0) \setminus \{u_0\}}^{v_0} = r_{\text{pa}(v_0)}^{v_0}$ if and only if $r_{\text{pa}(v_0) \setminus \{u_0\}}^{v_0} \neq r_{\text{pa}(v_0)}^{v_0} - 1$. $\square$

Lemma A.4 (Jutten and Comon [2010]). *Let $\varepsilon = (\varepsilon_1, \dots, \varepsilon_p)$ be any p -variate random vector, and $A \in \mathbb{R}^{s \times p}$ for any $s \in \mathbb{N}$, then*

$$\begin{aligned} \mathcal{C}^{(k)}(A\varepsilon)_{i_1, \dots, i_k} &= \sum_{1 \leq j_1, \dots, j_k \leq p} \mathcal{C}^{(k)}(\varepsilon)_{j_1, \dots, j_k} a_{j_1, i_1} \cdots a_{j_k, i_k} \\ &= (\mathcal{C}^{(k)}(\varepsilon) \bullet_k A)_{i_1, \dots, i_k}, \end{aligned}$$

where $\bullet_k$ denotes the Tucker product.

Proof for Lemma 3.3.

Proof. Let $\mathcal{C}_J^{(k)}$ denote a k -way cumulant tensor with an index set $J = \{j_1, \dots, j_k\}$. Let $U(J)$ be the number of unique elements in a multiset J . Let $[p] = \{1, \dots, p\}$, and let $\mathbf{1}\{\cdot\}$ be the a function which takes the value 1 if the condition is fulfilled and 0 otherwise. We can then write

$$\begin{aligned} \dim(\mathcal{T}^{(k)}(G_B)) &= \sum_{\substack{j_1, \dots, j_k \in [p] \\ \mathcal{C}_J \in \text{Sym}_k(\mathbb{R}^p)}} \mathbf{1}\{\mathcal{C}_J^{(k)} \neq 0\} \\ &= \sum_{i=1}^k \sum_{\substack{j_1, \dots, j_k \in [p] \\ \mathcal{C}_J \in \text{Sym}_k(\mathbb{R}^p) \\ U(J)=i}} \mathbf{1}\{\mathcal{C}_J^{(k)} \neq 0\} \\ &= \sum_{i=1}^k n_i(G_B) \sum_{\substack{j_1, \dots, j_k \in [i] \\ \mathcal{C}_J \in \text{Sym}_k(\mathbb{R}^i) \\ U(J)=i}} \mathbf{1} \\ &= \sum_{i=1}^{\min(p, k)} n_i(G_B) \cdot \binom{k-1}{i-1}. \end{aligned}$$

The last equality follows from the fact that the binomial coefficient $\binom{k-1}{i-1}$ gives the number of size k multisets with their elements chosen from a size i set, where we only allow the multisets to be chosen in the way that all of the i elements are chosen at least once. Note that $k \geq \min\{p, k\} \geq i$, so the binomial coefficient is always defined. $\square$

Proof for Theorem 3.4.

Proof. Let us consider two arbitrary ADMGs G and $\tilde{G}$ with $\mathcal{M}(G_1) = \mathcal{M}(\tilde{G})$, i.e. they are model equivalent. Then especially

$$\dim(\mathcal{T}^{(k)}(G)) - \dim(\mathcal{T}^{(k)}(\tilde{G})) = 0 \quad (\text{A.4})$$

for all $k \in \mathbb{N}$. Recall that we define the cumulant tensor space for arbitrary G to be

$$\mathcal{T}^{(k)}(G) = \{\mathcal{C} \bullet_k B_\Lambda \mid \mathcal{C} \in \mathcal{T}^{(k)}(G_B), \Lambda \in \mathbb{R}^{G_D}\},$$

where $\bullet_k$ denotes the Tucker product. This means that we can parametrize any $\hat{\mathcal{C}} \in \mathcal{T}^{(k)}(G)$ using $\mathcal{C} \in \mathcal{T}^{(k)}(G_B)$ and $\Lambda \in \mathbb{R}^{G_D}$. So we conclude that

$$\dim(\mathcal{T}^{(k)}(G)) \leq \dim(\mathcal{T}^{(k)}(G_B)) + \dim(\mathbb{R}^{G_D})$$

for any $k \in \mathbb{N}$. From the inequality above we obtain that

$$0 \leq \dim(\mathcal{T}^{(k)}(G)) - \dim(\mathcal{T}^{(k)}(G_B)) \leq c, \quad (\text{A.5})$$

and similarly

$$0 \leq \dim(\mathcal{T}^{(k)}(\tilde{G})) - \dim(\mathcal{T}^{(k)}(\tilde{G}_B)) \leq c, \quad (\text{A.6})$$

where $c = \max \{ \dim(\mathbb{R}^{G_D}), \dim(\mathbb{R}^{\tilde{G}_D}) \}$ serves as a constant upper bound that does not depend on k . So using Equation (A.5) and Equation (A.6) we get

$$\left| \left(\dim(\mathcal{T}^{(k)}(G)) - \dim(\mathcal{T}^{(k)}(G_B)) \right) - \left(\dim(\mathcal{T}^{(k)}(\tilde{G})) - \dim(\mathcal{T}^{(k)}(\tilde{G}_B)) \right) \right| \leq c.$$

Using Equation (A.4), the inequality above simplifies to

$$| \dim(\mathcal{T}^{(k)}(G_B)) - \dim(\mathcal{T}^{(k)}(\tilde{G}_B)) | \leq c,$$

whenever G_1 and G_2 are model equivalent. The last inequality together with Lemma 3.3 suggest that

$$\left| \underbrace{\sum_{i=1}^{\min(p,k)} (n_i(G_B) - n_i(\tilde{G}_B)) \cdot \binom{k-1}{i-1}}_{:=g(k)} \right| \leq c. \quad (\text{A.7})$$

We are interested in higher moments, so we will from now on look at $k \geq p$. We now define $d_i := n_i(G_B) - n_i(\tilde{G}_B)$, then $g(k)$ in Equation (A.7) further simplifies to

$$\begin{aligned} g(k) &= \sum_{i=2}^p \frac{d_i}{(i-1)!} \frac{(k-1)!}{(k-i)!} \\ &= \sum_{i=2}^p \frac{d_i}{(i-1)!} (k-1)(k-2) \cdots (k-i+1). \end{aligned}$$

Clearly $g(k)$ can be seen as a polynomial function of k since d_i and p are fixed for two given ADMGs G and $\tilde{G}$.

If G and $\tilde{G}$ are model equivalent, we must have

$$|g(k)| \leq c \quad \text{for all } k \geq p, k \in \mathbb{N}. \quad (\text{A.8})$$

Consider the asymptotic behavior of $g(k)$ for k large enough. If $d_p \neq 0$, then

$$g(k) \sim \frac{d_p}{(p-1)!} k^{p-1}$$

asymptotically. Thus there exists some k large enough so that $|g(k)| > c$, which leads to a contradiction to Equation (A.8). So we conclude that $d_p = 0$. Setting d_p to zero, we then have

$$g(k) \sim \frac{d_{p-1}}{(p-2)!} k^{p-2}$$

asymptotically if $d_{p-1} \neq 0$. Similarly, we argue that $d_{p-1} = 0$ must hold as well. Continuing with the same argument, we can show that $d_i = 0$ for all $2 \leq i \leq p$, $i \in \mathbb{N}$. We conclude the proof by noticing that $d_1 = 0$ and $n_i(G_B) = n_i(\tilde{G}_B) = 0$ for $i > p$. $\square$

Proof for Corollary 3.5.

Proof. By definition we have $n_2(G_B) = |E_{\leftrightarrow}|$, so the first equality follows. For $|D_{\max}| = |\tilde{D}_{\max}|$, note that

$$|D_{\max}| = \max \{ i \in \mathbb{N}^+ : n_i(G_B) \neq 0 \}.$$

The dimensional equality $\dim(\mathcal{T}^{(k)}(G_B)) = \dim(\mathcal{T}^{(k)}(\tilde{G}_B))$ for all $k \in \mathbb{N}^+$ follows directly by applying Lemma 3.3 and Theorem 3.4. $\square$

Example A.1 (The four-cycle is identifiable.). Consider a four-cycle $G = G_B$ as in Figure 6. We show that there exists no ADMG $\tilde{G}$ which satisfies the condition in Theorem 3.4 such that $\mathcal{M}(\tilde{G}) \subseteq \mathcal{M}(G)$. First note that $(I - \Lambda)^T B_{\tilde{\Lambda}} = B_{\tilde{\Lambda}}$. For model inclusion, since $v_1 \leftrightarrow v_3 \notin G$ and $v_2 \leftrightarrow v_4 \notin G$, by Theorem 3.1, we require that $(B_{\tilde{\Lambda}})_{1w_1} \cdot (B_{\tilde{\Lambda}})_{3w_2} = 0$ and $(B_{\tilde{\Lambda}})_{2w_1} \cdot (B_{\tilde{\Lambda}})_{4w_2} = 0$ hold for all $w_1 = w_2$ and $w_1 \leftrightarrow w_2 \in \tilde{G}$. Considering the condition in Theorem 3.4, the bidirected part of $\tilde{G}$ must also be a four-cycle. However, the other two four cycles 1-3-2-4-1 and 1-2-4-3-1 are not allowed because the bidirected edge $v_1 \leftrightarrow v_3 \in \tilde{G}$ or $v_2 \leftrightarrow v_4 \in \tilde{G}$ would imply $(B_{\tilde{\Lambda}})_{11} \cdot (B_{\tilde{\Lambda}})_{33} = 1$ or $(B_{\tilde{\Lambda}})_{22} \cdot (B_{\tilde{\Lambda}})_{44} = 1$. So $\tilde{G}_B$ must be the four-cycle 1-2-3-4-1. Now substituting all $w_1 = w_2$ and $w_1 \leftrightarrow w_2 \in \tilde{G}$ into $(B_{\tilde{\Lambda}})_{1w_1} \cdot (B_{\tilde{\Lambda}})_{3w_2} = 0$ and $(B_{\tilde{\Lambda}})_{2w_1} \cdot (B_{\tilde{\Lambda}})_{4w_2} = 0$, and noting that $(B_{\tilde{\Lambda}})_{ii} = 1$ on the diagonal, we conclude that $B_{\tilde{\Lambda}} = I$, which means $\tilde{G} = G$.

B ADDITIONAL FIGURES

The average CPU time increases exponentially with the number of variables p as suggested by the red dashed line in Figure 9, where the time axis is represented on a log scale.

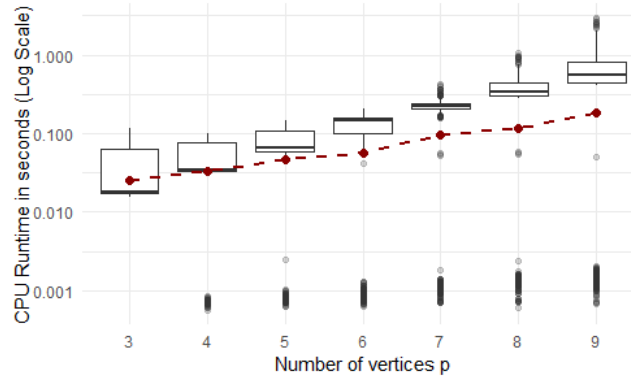
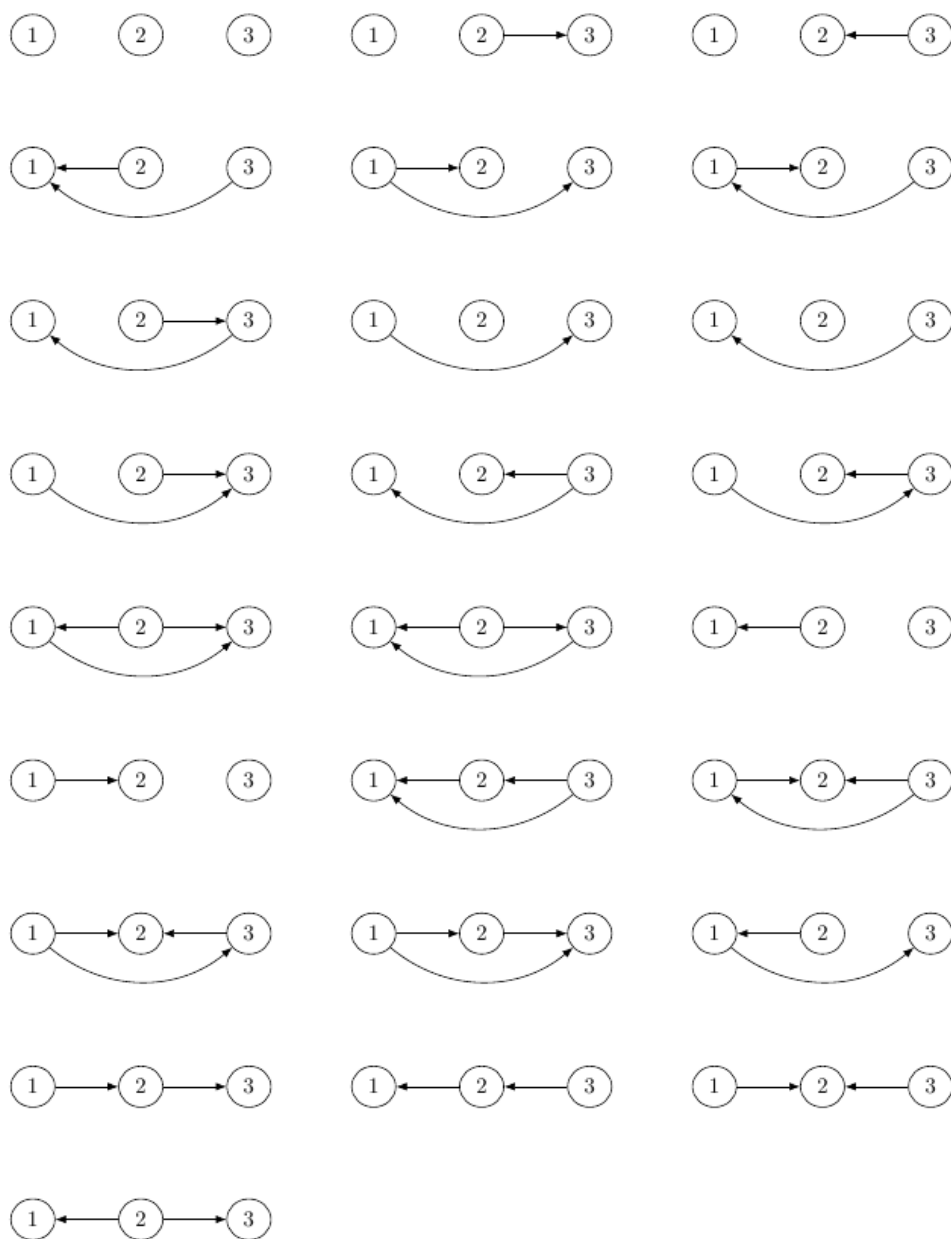


Figure 9: CPU time in log scale using Algorithm 2. The Red dashed line represents the average CPU time.

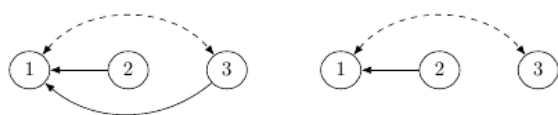
C EQUIVALENCE CLASS SEPARATION FOR THREE NODES

There are in total $25 \times 2^{\binom{3}{2}} = 200$ ADMGs with three vertices. These are divided into 74 equivalence classes. Classes 1–25 are trivial classes, each containing exactly one ADMG. In other words, each of the 25 DAGs with no bidirected edges forms its own equivalence class. Another trivial class is the final one, which contains all 25 ADMGs whose bidirected part is fully connected.





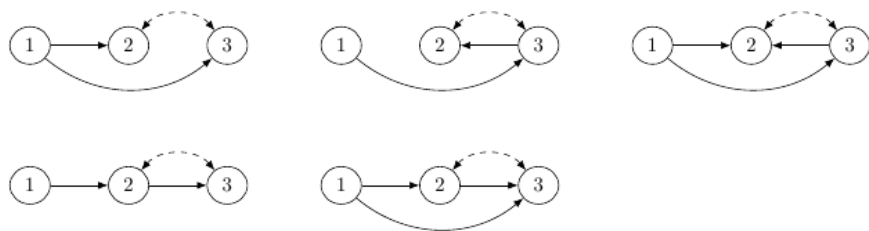
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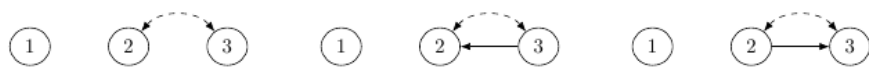
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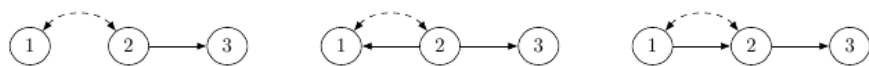
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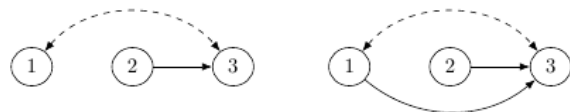
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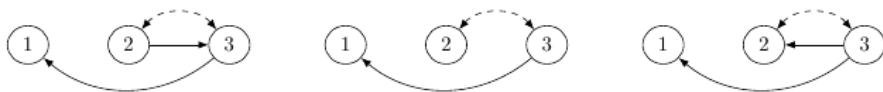
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Class 31



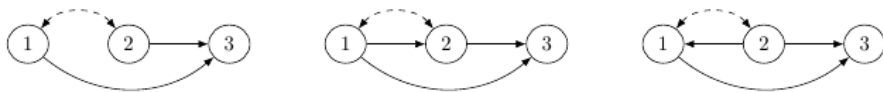
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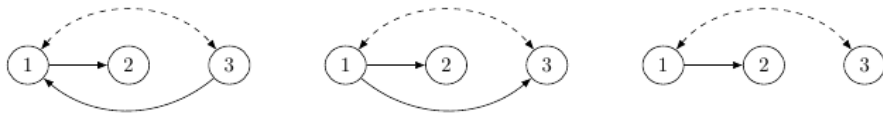
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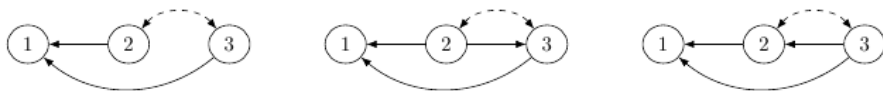
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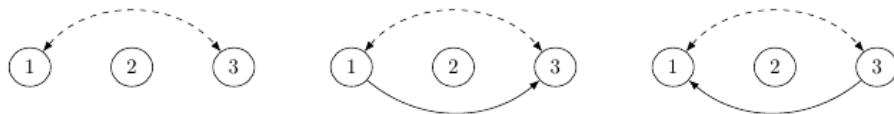
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Class 36



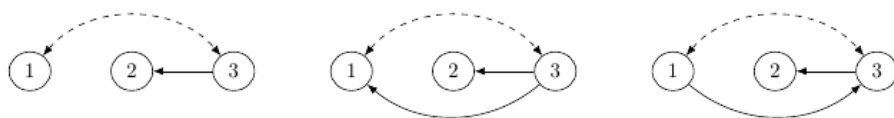
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Class 38



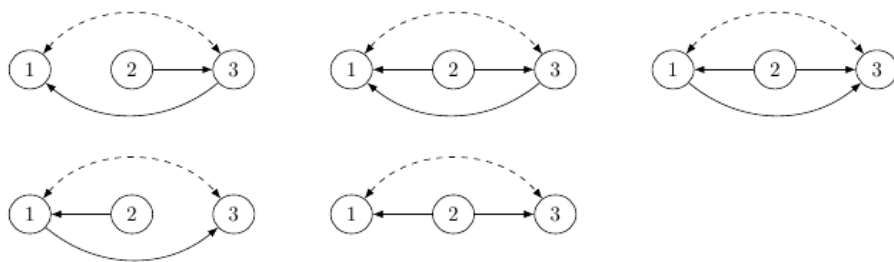
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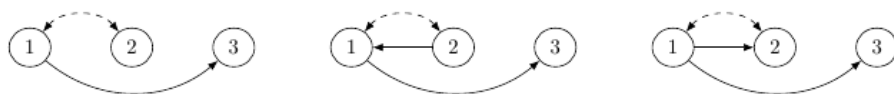
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Class 41



Class 42



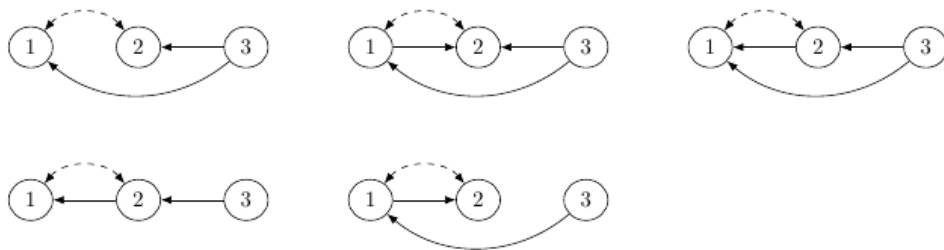
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Class 44



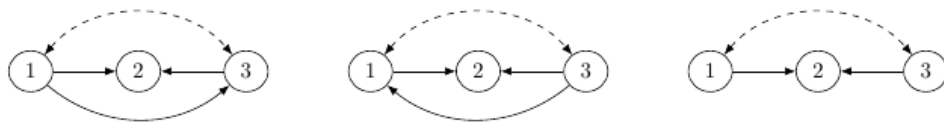
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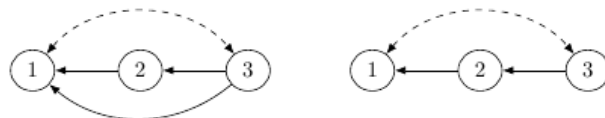
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Class 47



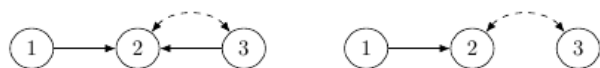
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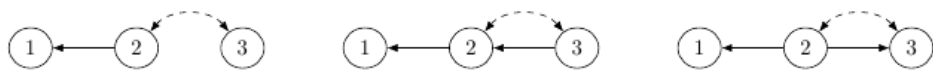
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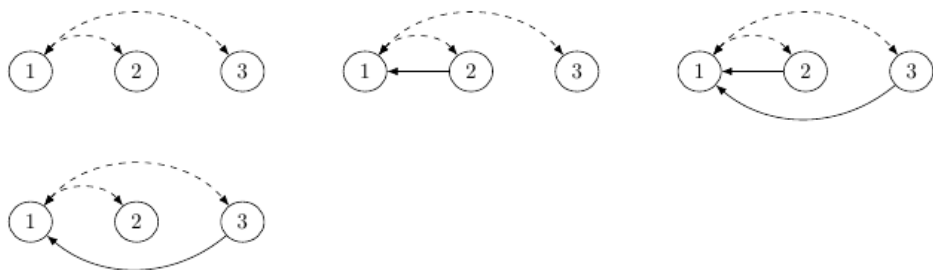
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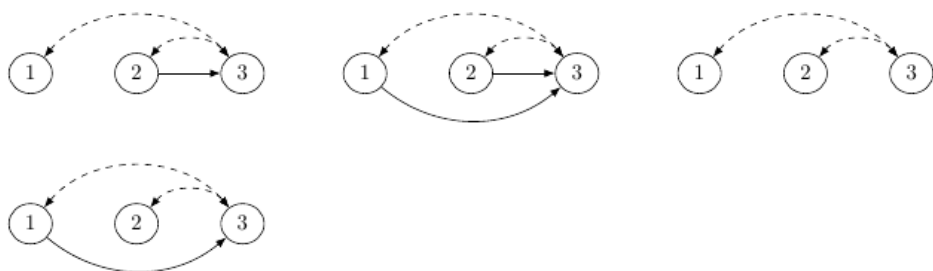
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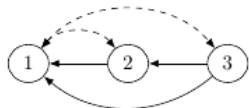
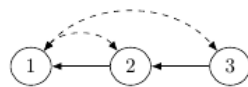
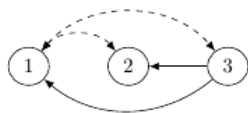
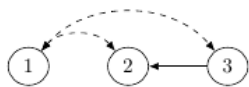
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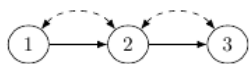
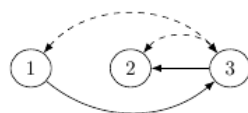
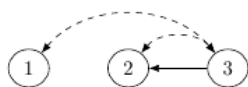
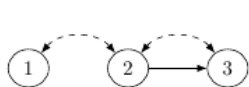
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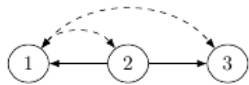
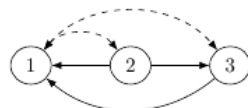
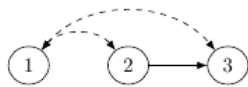
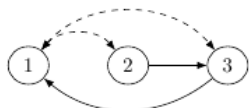
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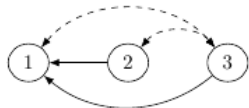
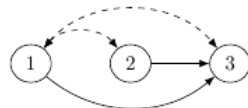
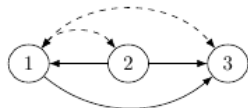
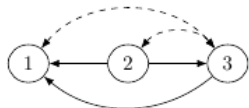
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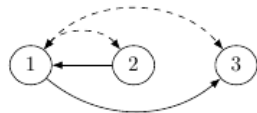
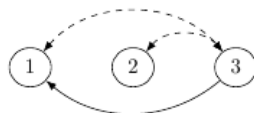
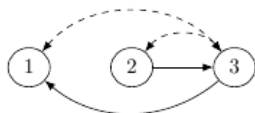
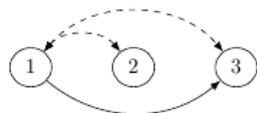
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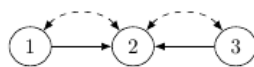
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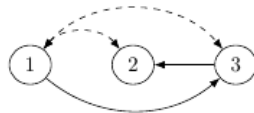
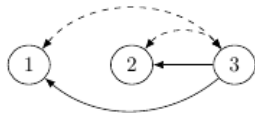
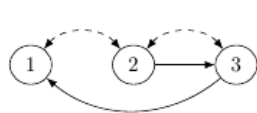
Class 58



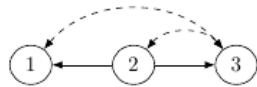
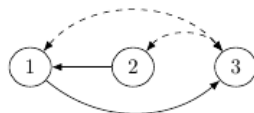
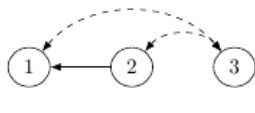
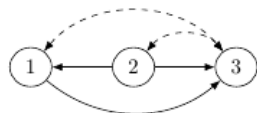
Class 59



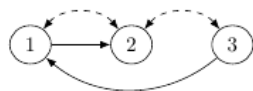
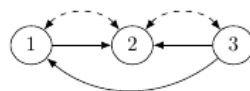
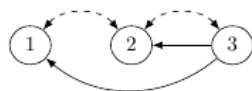
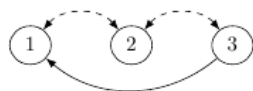
Class 60



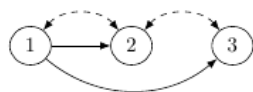
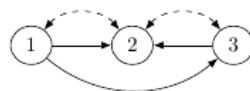
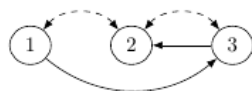
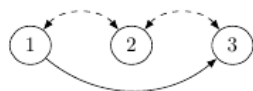
Class 61



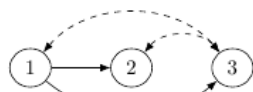
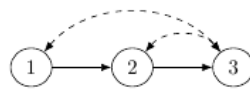
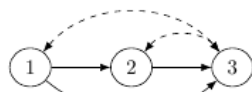
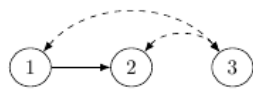
Class 62



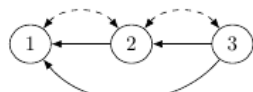
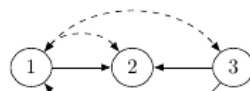
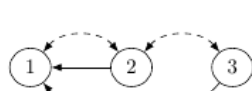
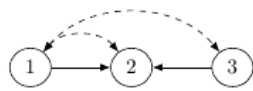
Class 63



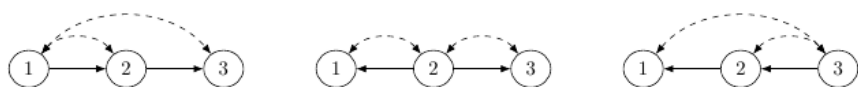
Class 64



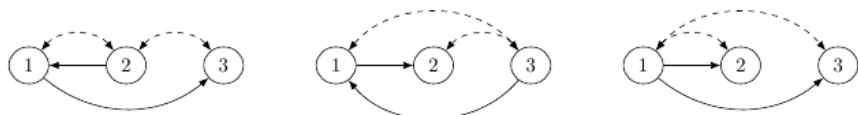
Class 65



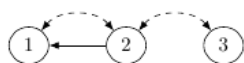
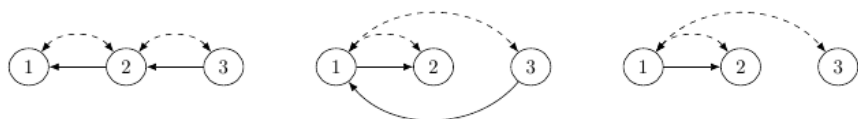
Class 66



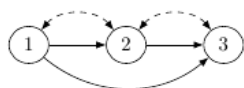
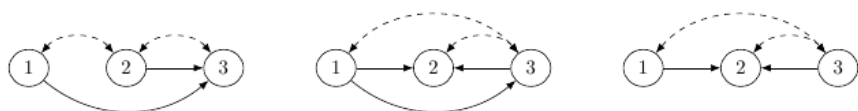
Class 67



Class 68



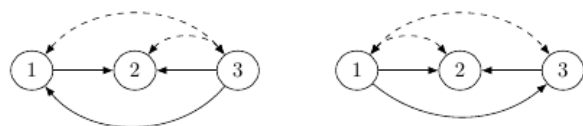
Class 69



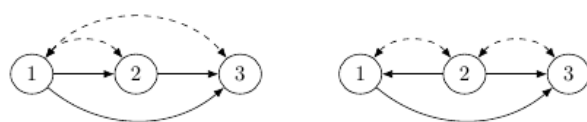
Class 70



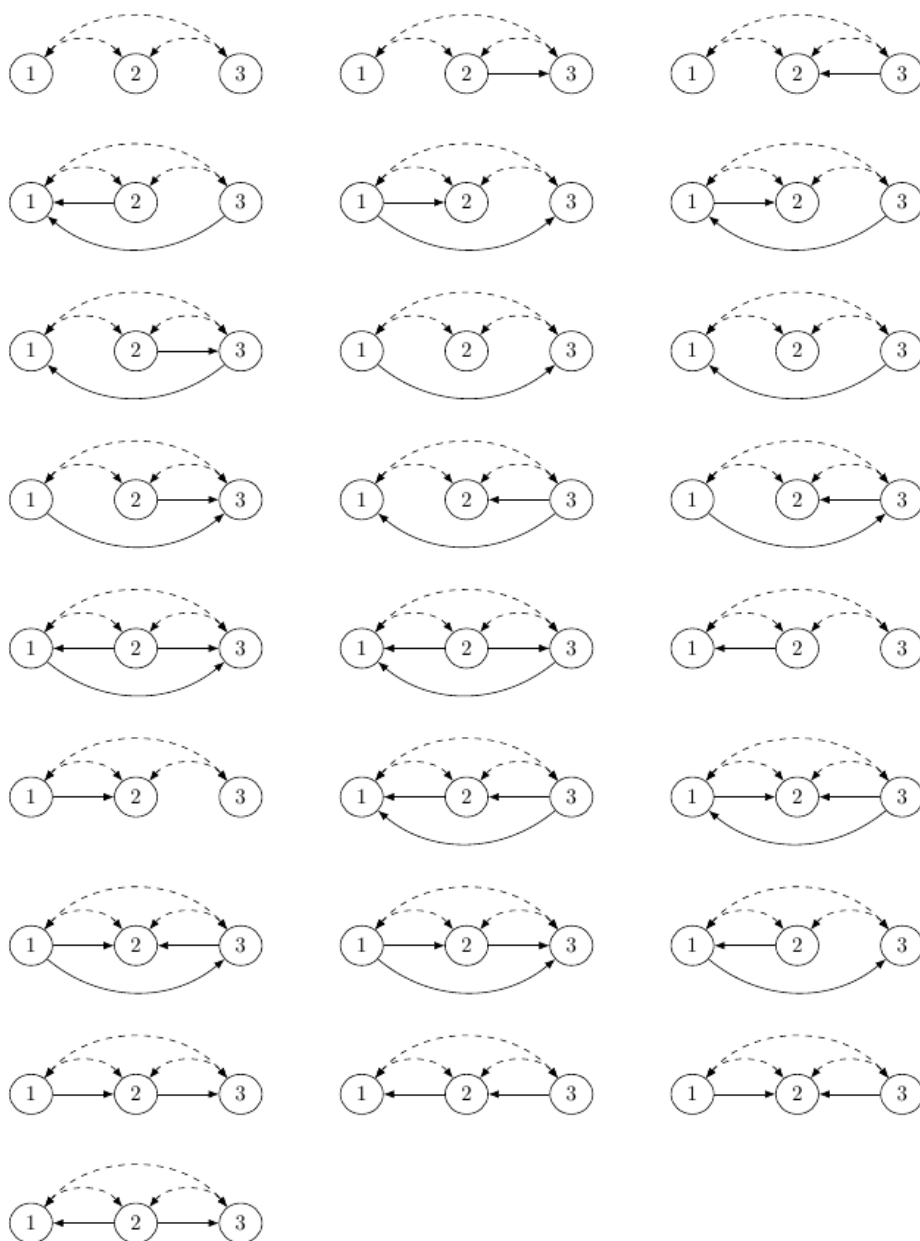
Class 71



Class 72



Class 73



Classes 74